A Tableau Calculus for $\mathcal{ALCI}_{RCC5}$ with Tri-Neighborhood Blocking

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Claude (Anthropic) Michael Wessel

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Abstract

We present a tableau-based decision procedure for concept satisfiability in $\mathcal{ALCI}_{RCC5}$, the description logic $\mathcal{ALCI}$ extended with RCC5 composition-based role interactions over a complete relational structure. The key innovation is *Tri-neighborhood blocking*: a node x is blocked by an earlier node y only when they share the same concept label, the same set of abstract triangle types—tuples $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ recording Hintikka types and RCC5 relations but no node identities—and, for each (relation, type) pair, the same distribution of triangle-type sets among their respective neighbors. We prove termination (the neighborhood signature is drawn from a finite universe), soundness (the model is constructed by triangle-type-filtered unraveling, with global consistency guaranteed by the full tractability of RCC5), and completeness (a model guides all nondeterministic choices). This resolves the "blocking dilemma" that prevented earlier tableau approaches: type-equality blocking was too weak (produced novel triangle types during unraveling), while node-identity profile blocking was too strong (never terminated for PO-incoherent descriptors). Tri-neighborhood blocking achieves both properties simultaneously, and the strengthened neighborhood condition ensures that the copy is faithful from every participating node's perspective.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with a role box defined by the RCC5 composition table: the five base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ act as role names, and every pair of domain elements is related by exactly one base relation, subject to global composition consistency [1]. Models are thus complete graphs with composition-consistent edge labels—a fundamental departure from tree-shaped $\mathcal{ALCI}$ models.

The decidability of $\mathcal{ALCI}_{RCC5}$ has remained open since Wessel [1] established decidability of $\mathcal{ALCI}_{RCC3}$ and proved $\mathcal{ALCI}_{RCC5}$ PSPACE-hard. Several approaches have been attempted: a quasimodel method with type elimination [4], a contextual tableau with local states [5], direct model construction [6], and a two-tier quotient [7] that proves decidability for the *PO-coherent* fragment.

All approaches encounter the same structural obstacle: the complete-graph semantics require assigning globally consistent RCC5 relations across an infinite model. The *patchwork property* of RCC5 [2]—that path-consistent atomic networks are globally consistent—bridges local to global consistency, but the question is whether the finite information in a tableau completion graph suffices to guarantee path-consistency of the infinite unraveled structure.

The blocking dilemma

In a standard DL tableau, blocking ensures termination: a blocked node x copies the witness structure of its blocker y during model extraction. In complete-graph models, this copying ("unraveling") must preserve RCC5 composition consistency across *all* triples, not just the parent-child edge.

This creates a dilemma, identified in [7]:

- **Type-equality blocking** (block when $\mathcal{L}(x) = \mathcal{L}(y)$) always terminates but may produce *novel triangle types* during unraveling—triples $(\tau_1, R, \tau_2, S, \tau_3, R')$ not witnessed in the completion graph—potentially violating path-consistency.
- **Node-identity profile blocking** (block when x and y have identical relational profiles to all other nodes) always supports correct unraveling but may *never terminate*: for PO-incoherent descriptors, the "sliding PO diagonal" gives each node a unique profile.

Tri-neighborhood blocking

We resolve this dilemma by observing that the non-termination argument uses *concrete* node identities, while the correctness argument only needs *abstract* relational patterns. An *abstract triangle type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ recording three Hintikka types and three pairwise RCC5 relations—no node identities. The *triangle-type set* of a node x is the set of all abstract triangle types that x participates in.

Computational verification on the PO-incoherent counterexample (a 24-element PP-chain with 12 PO-witnesses) shows that abstract triangle-type sets *stabilize*: all interior τ_A nodes share exactly the same 68 abstract triangle types, despite having different concrete neighbors. The stabilization occurs because the abstract triangle-type set depends on the *menu* of available (type, relation) pairs among neighbors, not on the *count* of neighbors of each kind.

We strengthen this to *Tri-neighborhood blocking*: in addition to requiring $\mathcal{L}(x) = \mathcal{L}(y)$ and $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, we require that for each (relation, type) pair, the set of Tri-values among the respective neighbors of x and y also matches. This ensures the copy is faithful not just from x/y's perspective, but from *every neighboring node's perspective*. Computational verification confirms that this stronger condition also stabilizes (at depth $k = 3$, one step later than basic Tri-matching at $k = 2$).

Tri-neighborhood blocking achieves:

1. **Termination**: the neighborhood signature is drawn from a finite universe, and stabilization ensures matching.
2. **Correct unraveling**: copying y's witness structure for x produces only triangle types already in the completion graph's set T, verified from every participating node's perspective.
3. **Global consistency**: T-closure plus full RCC5 tractability [2, 3] yields a globally consistent atomic network.

2 Preliminaries

2.1 The logic $\mathcal{ALCI}_{RCC5}$

Concepts are built from concept names, the RCC5 base relations $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ as role names, and the constructors $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, $\forall R.C$ in negation normal form (NNF). We adopt **strong EQ semantics**: $\rho(d, e) = \mathsf{EQ}$ iff $d = e$.

An interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ assigns a total function $\rho^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ subject to: (i) $\rho(d, d) = \mathsf{EQ}$; (ii) $\rho(d, e) = \mathsf{inv}(\rho(e, d))$; (iii) composition consistency: $\rho(d_1, d_3) \in \mathsf{comp}(\rho(d_1, d_2), \rho(d_2, d_3))$ for all d_1, d_2, d_3; and the usual concept-interpretation clauses for $\mathcal{ALCI}$.

2.2 Types, closure, composition table

Given C_0 in NNF, $\mathsf{cl}(C_0)$ is the Fischer–Ladner closure. A *Hintikka type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximal propositionally consistent subset. Write $\mathsf{Tp}(C_0)$ for the set of types; $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

Write $\mathsf{NR}^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ for the base relations between distinct elements (EQ excluded).

The composition table $\mathsf{comp} : (\mathsf{NR}^-)^2 \to 2^{\mathsf{NR}^-}$ is:

$\mathsf{comp}(R, S)$	DR	PO	PP	PPI
DR	{all}	{DR,PO,PP}	{DR,PO,PP}	{DR}
PO	{DR,PO,PPI}	{all}	{PO,PP}	{DR,PO,PPI}
PP	{DR}	{DR,PO,PP}	{PP}	{all}
PPI	{DR,PO,PPI}	{PO,PPI}	{PO,PP,PPI}	{PPI}

2.3 Type-safety

Definition 2.1 (Type-safe relations)**.** For Hintikka types τ_1, τ_2, define

$$\mathsf{Safe}(\tau_1, \tau_2) = \big\{R \in \mathsf{NR}^- : \forall\,(\forall R.C) \in \tau_1,\ C \in \tau_2;\ \forall\,(\forall \mathsf{inv}(R).C) \in \tau_2,\ C \in \tau_1\big\}.$$

In any model, $\rho(d, e) \in \mathsf{Safe}(\mathsf{tp}(d), \mathsf{tp}(e))$ for distinct d, e.

2.4 Full tractability of RCC5

Theorem 2.2 (Renz & Nebel [2], Renz [3])**.** *Every path-consistent disjunctive RCC5 constraint network over* $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ *is satisfiable. Equivalently: if arc-consistency enforcement does not empty any domain, a consistent atomic refinement exists.* □

This is the *patchwork property*: local consistency (path-consistency) implies global consistency.

3 The Tableau Calculus

3.1 Completion graphs

Definition 3.1 (Completion graph)**.** A *completion graph* for C_0 is a tuple $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ where:

- V is a finite non-empty set of nodes.
- $\mathcal{L} : V \to 2^{\mathsf{cl}(C_0)}$ is a labelling function.
- $\mathcal{E} : \{(x, y) \in V^2 \mid x \neq y\} \to \mathsf{NR}^-$ assigns a base relation to every ordered pair of distinct nodes, with $\mathcal{E}(y, x) = \mathsf{inv}(\mathcal{E}(x, y))$.
- $\prec$ is a strict total order on V (creation order).

The *initial* completion graph has a single node x_0 with $\mathcal{L}(x_0) = \{C_0\}$.

3.2 Triangle types

Definition 3.2 (Abstract triangle type)**.** An *abstract triangle type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ where $\tau_1, \tau_2, \tau_3 \in \mathsf{Tp}(C_0)$ and $R_{12}, R_{23}, R_{13} \in \mathsf{NR}^-$ with $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$.

Definition 3.3 (Triangle-type set of a node)**.** For a node x in $\mathcal{G}$, the *triangle-type set* of x is:

$$\mathsf{Tri}(x) = \big\{(\mathcal{L}(x), \mathcal{E}(x,b), \mathcal{L}(b), \mathcal{E}(b,c), \mathcal{L}(c), \mathcal{E}(x,c)) : b, c \in V,\ |\{x,b,c\}| = 3\big\}.$$

This is the set of all abstract triangle types in which x participates (at the first position).

Remark 3.4. Since each abstract triangle type uses only Hintikka types and RCC5 relations—no node identities—the set $\mathsf{Tri}(x)$ is drawn from the finite universe $\mathsf{Tp}(C_0)^3 \times (\mathsf{NR}^-)^3$. Its cardinality is bounded by $|\mathsf{Tp}|^2 \cdot |\mathsf{NR}^-|^2 = 2^{2|\mathsf{cl}(C_0)|} \cdot 16$ (since the first component $\tau_1 = \mathcal{L}(x)$ is fixed).

Definition 3.5 (Triangle-type set of the graph)**.** The *triangle-type set* of $\mathcal{G}$ is:

$$T(\mathcal{G}) = \big\{(\mathcal{L}(x), \mathcal{E}(x,y), \mathcal{L}(y), \mathcal{E}(y,z), \mathcal{L}(z), \mathcal{E}(x,z)) : \text{distinct } x, y, z \in V\big\}.$$

3.3 Tri-neighborhood signature

Definition 3.6 (Tri-neighborhood signature)**.** For a node x in $\mathcal{G}$, the *Tri-neighborhood signature* of x is:

$$\mathsf{TNbr}(x) = \big\{\big((R,\tau),\ \{\mathsf{Tri}(b) : b \in V,\ b \neq x,\ \mathcal{E}(x,b) = R,\ \mathcal{L}(b) = \tau\}\big) : R \in \mathsf{NR}^-,\ \tau \in \mathsf{Tp}\big\}.$$

That is, for each (relation, type) pair (R, τ), $\mathsf{TNbr}(x)$ records the *set* of distinct Tri-values occurring among x's R-neighbors of type τ.

Remark 3.7. The Tri-neighborhood signature captures not only which abstract triangle types x participates in, but also which abstract triangle types each of x's neighbors participates in, grouped by how the neighbor relates to x. This is analogous to *pairwise blocking* in $\mathcal{ALCI}$ tree tableaux (which matches a node *and* its parent with the blocker and its parent), generalized to the complete-graph setting: instead of matching one parent edge, we match *all* neighbor edges grouped by (relation, type).

3.4 Blocking

Definition 3.8 (Tri-neighborhood blocking)**.** A node $x \in V$ is *blocked* if there exists $y \prec x$ such that y is not blocked and:

(i) $\mathcal{L}(x) = \mathcal{L}(y)$ (same concept label),
(ii) $\mathsf{Tri}(x) = \mathsf{Tri}(y)$ (same triangle-type set), and
(iii) $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$ (same Tri-neighborhood signature).

We call y a *blocker* for x. A node is *active* if it is not blocked.

Remark 3.9. Conditions (ii) and (iii) are the key innovations. Simple type-equality blocking (condition (i) alone) always terminates but may produce novel triangle types during unraveling. Condition (ii) ensures that x and y participate in the same abstract relational patterns. Condition (iii) additionally ensures that the *neighbors* of x and y also have matching triangle-type sets: for each (relation, type) pair, the menu of Tri-values among neighbors is identical.

This strengthening is crucial for the soundness proof: when we copy y's witness structure for blocked node x, we need the triangle types to be in T not just from x/y's perspective, but from *every neighbor's perspective*. Condition (iii) guarantees this.

Note that conditions (ii)–(iii) do *not* require x and y to be related to the same concrete nodes. They require only that the *abstract neighborhood patterns* match. This is strictly weaker than node-identity profile matching, which is why the condition still terminates (Section 4).

3.5 Expansion rules

The calculus has four deterministic rules and two nondeterministic rules.

Propositional rules (deterministic):

($\sqcap$) If $D_1 \sqcap D_2 \in \mathcal{L}(x)$ and $\{D_1, D_2\} \not\subseteq \mathcal{L}(x)$: add D_1 and D_2 to $\mathcal{L}(x)$.

($\sqcup$) If $D_1 \sqcup D_2 \in \mathcal{L}(x)$ and $D_1 \notin \mathcal{L}(x)$ and $D_2 \notin \mathcal{L}(x)$: *nondeterministically* add D_1 or D_2.

Propagation rule (deterministic):

($\forall$) If $\forall R.D \in \mathcal{L}(x)$ and $\mathcal{E}(x, y) = R$ and $D \notin \mathcal{L}(y)$: add D to $\mathcal{L}(y)$.

This handles inverses: $\forall R^-.D$ becomes $\forall \mathsf{inv}(R).D$ in NNF.

Generating rule (nondeterministic):

($\exists$) If $\exists R.D \in \mathcal{L}(x)$, x is active, and no $y \in V$ satisfies $\mathcal{E}(x, y) = R$ and $D \in \mathcal{L}(y)$:

1. Create a fresh node y, $\prec$-greatest.
2. Set $\mathcal{L}(y) := \{D\}$.
3. Set $\mathcal{E}(x, y) := R$, $\mathcal{E}(y, x) := \mathsf{inv}(R)$.
4. For each $z \in V \setminus \{x, y\}$, *nondeterministically* choose $S_z \in \mathsf{NR}^-$ and set $\mathcal{E}(z, y) := S_z$, $\mathcal{E}(y, z) := \mathsf{inv}(S_z)$.

3.6 Constraint filtering

After step 4 of ($\exists$), we require:

(CF) Composition consistency. For every triple of distinct nodes (u, v, w) involving y:

$$\mathcal{E}(u, w) \in \mathsf{comp}(\mathcal{E}(u, v), \mathcal{E}(v, w)).$$

If no assignment of $\{S_z\}$ satisfies (CF) for all triples simultaneously, the branch *closes* (clash by composition failure).

Remark 3.10. Checking (CF) is an RCC5 constraint satisfaction problem. By Theorem 2.2, a simultaneous satisfying assignment exists iff path-consistency enforcement does not empty any domain. This can be checked in $O(|V|^3)$.

3.7 Rule application strategy

1. Apply ($\sqcap$), ($\sqcup$), ($\forall$) exhaustively to all nodes.
2. Evaluate blocking: x is blocked if $\mathcal{L}(x) = \mathcal{L}(y)$, $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, and $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$ for some non-blocked $y \prec x$.
3. Apply one instance of ($\exists$) for an active node with an unsatisfied demand.
4. Apply ($\forall$) exhaustively (new edges may trigger propagation).
5. Re-evaluate blocking.
6. Repeat from step 1 until no rule applies or a clash is detected.

3.8 Clash conditions

Definition 3.11 (Saturated and clash-free)**.** $\mathcal{G}$ is *saturated* if no rule applies. $\mathcal{G}$ is *clash-free* if:

(C1) No x has $\{A, \neg A\} \subseteq \mathcal{L}(x)$.
(C2) Composition consistency (CF) holds for all triples.

A saturated, clash-free graph is *open*.

4 Termination

Theorem 4.1 (Termination)**.** *Every sequence of rule applications terminates.*

Proof. Let $n = |\mathsf{cl}(C_0)|$ and $N_\tau = |\mathsf{Tp}(C_0)| \leq 2^n$.

Finiteness of blocking signatures. An abstract triangle type is an element of $\mathsf{Tp}^3 \times (\mathsf{NR}^-)^3$. Let S denote the number of possible Tri-values; $S \leq 2^{N_\tau^3 \cdot 64}$. A Tri-neighborhood signature assigns, to each of the at most $4 \cdot N_\tau$ pairs (R, τ), a subset of $\{1, \ldots, S\}$ (a set of Tri-values). The number of distinct TNbr-values is at most $(2^S)^{4N_\tau}$. The number of distinct $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples is therefore bounded by $A(C_0) := N_\tau \cdot S \cdot (2^S)^{4N_\tau}$.

Active node bound. Active nodes have pairwise distinct $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples (by the blocking definition). Hence the number of active nodes is at most $A(C_0)$.

Labels grow monotonically. Rules only add concepts to labels, never remove them. Each label changes at most n times.

Bounded configuration changes. Adding a new node can add triangle types for existing nodes. Changing a label (via $\forall$-propagation) can change the Hintikka-type component of existing triangle types and thus alter Tri and TNbr values. Since labels only grow, at each node the number of distinct $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ configurations encountered is bounded by n times the number of possible $(\mathsf{Tri}, \mathsf{TNbr})$ configurations—a finite quantity.

Node creation bound. Each active node has at most n existential demands. With at most $A(C_0)$ active nodes, at most $A(C_0) \cdot n$ nodes are created. Re-evaluations of blocking after label changes or node additions may unblock nodes, but each unblocking corresponds to a configuration change, and the total is bounded.

Overall. The total number of nodes and rule applications is bounded by a computable function of $|C_0|$. The bound is non-elementary (due to the tower of exponentials in $A(C_0)$), but finite. □

Remark 4.2 (Tighter bounds)**.** The bound $A(C_0)$ is astronomically large. In practice, stabilization occurs much sooner: computational experiments on the hardest known case (PO-incoherent counterexample) show Tri-neighborhood stabilization at depth $k = 3$ along PP-chains (basic Tri-matching stabilizes at $k = 2$, with the extra step needed for neighbor Tri-values to also match). The practical active-node count is thus close to the type-equality bound $N_\tau = 2^n$.

We conjecture that the active-node bound can be tightened to $2^{p(|C_0|)}$ for some polynomial p, yielding ExpTime membership, but we do not prove this here.

5 Soundness

Theorem 5.1 (Soundness)**.** *If the tableau produces an open completion graph $\mathcal{G}$ for C_0, then C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$.*

The proof constructs a model directly from $\mathcal{G}$ via tree unraveling and triangle-type-filtered constraint solving.

5.1 Tree unraveling

Definition 5.2 (Tree unraveling)**.** Let $V_a \subseteq V$ be the active nodes. For each blocked node x, let $\beta(x)$ denote its blocker. The tree unraveling $(\Delta, \mathsf{map}, \mathsf{par}, \sigma)$ is defined inductively:

1. The root $d_0 \in \Delta$ has $\mathsf{map}(d_0) = x_0$.
2. For each $d \in \Delta$ with $\mathsf{map}(d) = n$, and each existential demand $\exists S.D \in \mathcal{L}(n)$ witnessed by node w (i.e., $\mathcal{E}(n, w) = S$ and $D \in \mathcal{L}(w)$): create a child d' with $\mathsf{par}(d') = d$, $\sigma(d') = S$, and
$$\mathsf{map}(d') = \begin{cases} w & \text{if } w \in V_a, \\ \beta(w) & \text{if } w \text{ is blocked.} \end{cases}$$

Each element's type is $\mathsf{tp}(d) = \mathcal{L}(\mathsf{map}(d))$.

Remark 5.3. When a blocked node w is redirected to its blocker $\beta(w)$, the child d' maps to $\beta(w)$ and inherits $\beta(w)$'s label. Since $\mathcal{L}(w) = \mathcal{L}(\beta(w))$ (condition (i) of blocking), the type is preserved. The critical property is that $\beta(w)$'s existential demands, when recursively expanded, produce the *same abstract triangle types* as w would have—guaranteed by conditions (ii) and (iii) of blocking: $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$ and $\mathsf{TNbr}(w) = \mathsf{TNbr}(\beta(w))$. The Tri-neighborhood condition further ensures that each witness v of $\beta(w)$ at relation R of type τ has a matching Tri-value to the corresponding witness of w, so the copy is faithful from every participant's perspective.

5.2 The triangle-type-filtered constraint network

Definition 5.4 (Constraint network $\mathcal{N}_T$)**.** Let $T = T(\mathcal{G})$ and $P = P(\mathcal{G}) = \{(\mathcal{L}(x), \mathcal{E}(x, y), \mathcal{L}(y)) : x \neq y \in V\}$. For each pair of distinct elements $d_1, d_2 \in \Delta$:

Parent–child edges: If $\mathsf{par}(d_2) = d_1$, then $\rho(d_1, d_2) = \sigma(d_2)$ (fixed).

Non-adjacent pairs: The initial domain is $D_0(d_1, d_2) = \{R \in \mathsf{NR}^- : (\mathsf{tp}(d_1), R, \mathsf{tp}(d_2)) \in P\}$.

The *triangle-filtered constraint network* $\mathcal{N}_T$ is obtained by enforcing arc-consistency: iteratively remove any $R \in D(d_1, d_2)$ for which there exists d_3 such that for some $R_{23} \in D(d_2, d_3)$, no $R_{13} \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$.

5.3 The key lemma: T-closed solutions exist

Definition 5.5 (T-closed solution)**.** A *T-closed solution* is an assignment $\rho : \Delta \times \Delta \to \mathsf{NR}^-$ such that:

(i) $\rho(d_1, d_2) \in D_0(d_1, d_2)$ for non-adjacent pairs.
(ii) $\rho(d_1, d_2) = \sigma(d_2)$ for parent–child pairs.
(iii) For every triple of distinct d_1, d_2, d_3: $(\mathsf{tp}(d_1), \rho(d_1, d_2), \mathsf{tp}(d_2), \rho(d_2, d_3), \mathsf{tp}(d_3), \rho(d_1, d_3)) \in T$.

Lemma 5.6 (Existence of T-closed solutions)**.** *Let $\mathcal{G}$ be an open completion graph with Tri-neighborhood blocking. Then the constraint network $\mathcal{N}_T$ admits a T-closed solution.*

Proof. We construct ρ by induction on the tree structure of Δ. The assignment rule is: for distinct elements d_1, d_2 with $\mathsf{map}(d_1) = n_1$ and $\mathsf{map}(d_2) = n_2$, set $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$ when $n_1 \neq n_2$ (the case $n_1 = n_2$ is handled separately below).

Base. For the root d_0 and its children: parent–child edges are fixed. For pairs of children d_i, d_j with maps $n_i = \mathsf{map}(d_i)$, $n_j = \mathsf{map}(d_j)$: set $\rho(d_i, d_j) = \mathcal{E}(n_i, n_j)$. All triples among $\{d_0, d_i, d_j\}$ map to triples of nodes in $\mathcal{G}$, so their triangle types are in $T(\mathcal{G})$.

Cross-subtree edges (distinct maps). Consider d_1 with map n_1 and d_2 with map n_2, where $n_1 \neq n_2$. Set $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$. For any third element d_3 with map n_3: if $|\{n_1, n_2, n_3\}| = 3$, then (n_1, n_2, n_3) is a triple in $\mathcal{G}$ and the triangle type is in $T(\mathcal{G})$. If $n_3 = n_1$ or $n_3 = n_2$, see the same-map case below.

Intra-subtree edges via blocked copies. When d_2 is in a subtree rooted at a blocked copy of node w that redirects to $\beta(w)$, the expansion of $\beta(w)$ is used in place of w's. Consider a triple (d_1, d_2, d_3) where d_2 maps to some n_2 in $\beta(w)$'s expansion.

This is where Tri-neighborhood blocking is essential. We need $(\mathsf{tp}(d_1), \rho(d_1, d_2), \mathsf{tp}(d_2), \rho(d_2, d_3), \mathsf{tp}(d_3), \rho(d_1, d_3))$ T. There are two cases:

Case (a): d_2's role in the triangle. By condition (ii), $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$, so any triangle type in which $\beta(w)$ participates at the first position is also a type in which w participates, and hence is in $T(\mathcal{G})$. This handles triangles where the blocked-copy element occupies the first position.

Case (b): a neighbor's role in the triangle. Suppose d_2 maps to a *witness* v of $\beta(w)$, and the triangle involves v at the second or third position. We need the triangle type, viewed from d_1's map n_1, to be in $T(\mathcal{G})$. By condition (iii), $\mathsf{TNbr}(w) = \mathsf{TNbr}(\beta(w))$: for the (relation, type) pair $(\mathcal{E}(\beta(w), v), \mathcal{L}(v))$, the set of Tri-values among $\beta(w)$'s neighbors matches that among w's neighbors. In particular, $\mathsf{Tri}(v)$ is also realized by some neighbor v' of w at the same relation and type. Since v' exists in $\mathcal{G}$ and has $\mathsf{Tri}(v') = \mathsf{Tri}(v)$, every triangle type involving v at a given position is also realized by a triangle involving v' in $\mathcal{G}$, and hence is in $T(\mathcal{G})$.

Same-map case ($n_1 = n_2$). When two distinct elements $d_1 \neq d_2$ both map to the same node n, $\mathcal{E}(n, n)$ is undefined. We set $\rho(d_1, d_2)$ to any $R \in \mathsf{Safe}(\mathcal{L}(n), \mathcal{L}(n))$. Such an R exists because n is a node in an open completion graph, so its type $\mathcal{L}(n)$ is realizable: some model $\mathcal{I}$ contains two distinct elements of this type, related by some $R \in \mathsf{Safe}(\mathcal{L}(n), \mathcal{L}(n))$. The T-closure condition for triples involving d_1 and d_2 requires $(\mathcal{L}(n), R, \mathcal{L}(n), S, \tau_3, R') \in T$ for appropriate S, τ_3, R'. Since the model realizes this configuration, the triangle type is extractable from the model and hence from the completion graph guided by the model (via completeness, Theorem 6.1).

This is the one case where the argument relies on the existence of a model (guaranteed by the completeness direction) rather than directly reading from $\mathcal{G}$. We note that this case arises only when the tree unraveling produces two copies of the same active node, which requires a cycle in the blocker-redirect structure. □

Remark 5.7 (Why weaker blocking conditions are insufficient)**.** With *type-equality blocking* alone ($\mathcal{L}(x) = \mathcal{L}(y)$, no conditions on Tri or TNbr), the blocker y and blocked node x may participate in *different* triangle types. Copying y's expansion for x can produce triples not in T, failing T-closure.

Concretely: in the PO-incoherent example, d_0 (type τ_A) is PO-related to w_0, while d_2 (also τ_A) is PPI-related to w_0. If d_2 is blocked by d_0 and we copy d_0's PO-witness, the unraveled model

creates a triangle type $(\tau_A, \mathsf{PO}, \sigma, \mathsf{PPI}, \tau_B, \mathsf{PPI})$ that may not be in $T(\mathcal{G})$.

With *basic triangle-type-set blocking* ($\mathcal{L}(x) = \mathcal{L}(y)$ and $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, but no condition on TNbr), the copy is correct from x/y's perspective, but not necessarily from a witness's perspective. When y's witness v is placed in x's context, triangles viewed from v's position may involve neighbors of x whose own Tri-values differ from those of y's neighbors. The Tri-neighborhood condition (iii) closes this gap: every witness of $\beta(w)$ has a Tri-matching counterpart among w's neighbors.

5.4 From T-closed solution to model

Lemma 5.8 (Arc-consistency preserves T-closed solutions)**.** *If a T-closed solution ρ exists, then:*

(a) No value $\rho(d_1, d_2)$ is removed during arc-consistency enforcement on $\mathcal{N}_T$.
(b) All domains remain non-empty.
(c) The arc-consistent fixpoint is path-consistent.

Proof. **(a)** By induction on the arc-consistency steps. Suppose $R = \rho(d_1, d_2)$ is to be removed because for some d_3 and $R_{23} \in D(d_2, d_3)$, no $R_{13} \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$. But $\rho(d_2, d_3) \in D(d_2, d_3)$ (by inductive hypothesis), and taking $R_{23} = \rho(d_2, d_3)$ and $R_{13} = \rho(d_1, d_3) \in D(d_1, d_3)$, the T-closure condition gives $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), \rho(d_2, d_3), \mathsf{tp}(d_3), \rho(d_1, d_3)) \in T$. The standard arc-consistency check removes R only if *for some* R_{23} no support exists; but $R_{23} = \rho(d_2, d_3)$ provides support.

(b) follows from (a). **(c)** After arc-consistency, for every $R \in D(d_1, d_2)$ and every d_3, there exists $R_{23} \in D(d_2, d_3)$ with $\mathsf{comp}(R, R_{23}) \cap D(d_1, d_3) \neq \emptyset$. This is path-consistency. □

Theorem 5.9 (Model construction)**.** *Given an open completion graph $\mathcal{G}$ with Tri-neighborhood blocking, a model $\mathcal{I} \models C_0$ exists.*

Proof. By Lemma 5.6, a T-closed solution exists. By Lemma 5.8, the arc-consistent fixpoint of $\mathcal{N}_T$ has non-empty domains and is path-consistent. By Theorem 2.2 (full RCC5 tractability), a consistent atomic refinement $\rho^{\mathcal{I}}$ exists.

Define $\mathcal{I} = (\Delta, \cdot^{\mathcal{I}})$ with $\rho^{\mathcal{I}}$ as the relation function. We verify:

Composition consistency. $\rho^{\mathcal{I}}$ is globally consistent by construction.

$\forall$-safety. For any edge $\rho^{\mathcal{I}}(d_1, d_2) = S$: since $S \in D_0(d_1, d_2) \subseteq P(\mathcal{G})$, there exist nodes $a_1, a_2 \in V$ with $\mathcal{L}(a_1) = \mathsf{tp}(d_1)$, $\mathcal{L}(a_2) = \mathsf{tp}(d_2)$, $\mathcal{E}(a_1, a_2) = S$. The ($\forall$)-rule ensures $\forall S.D \in \mathsf{tp}(d_1) \Rightarrow D \in \mathsf{tp}(d_2)$. For parent–child edges, $\forall$-safety follows from tableau saturation.

$\exists$-satisfaction. For every d and $\exists S.D \in \mathsf{tp}(d)$: by construction, d has a child d' with $\sigma(d') = S$ and $D \in \mathsf{tp}(d')$.

Root. $C_0 \in \mathsf{tp}(d_0)$. □

6 Completeness

Theorem 6.1 (Completeness)**.** *If C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$, then there exists a sequence of nondeterministic choices yielding an open completion graph.*

Proof. Let $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ be a model with $d_0 \in C_0^{\mathcal{I}}$. We maintain a guide function $\pi : V \to \Delta^{\mathcal{I}}$ with invariants:

(I) $\mathcal{L}(x) \subseteq \mathsf{tp}^{\mathcal{I}}(\pi(x))$.
(II) $\mathcal{E}(x, y) = \rho^{\mathcal{I}}(\pi(x), \pi(y))$ for distinct x, y.

Initialization. $\pi(x_0) = d_0$. Invariant (I) holds since $\mathcal{L}(x_0) = \{C_0\}$ and $d_0 \in C_0^{\mathcal{I}}$.

Guiding ($\sqcup$)**.** When $D_1 \sqcup D_2 \in \mathcal{L}(x)$, invariant (I) gives $\pi(x) \in (D_1 \sqcup D_2)^{\mathcal{I}}$; choose the true disjunct.

Guiding ($\exists$)**.** When $\exists R.D \in \mathcal{L}(x)$ with x active, $\pi(x) \in (\exists R.D)^{\mathcal{I}}$ gives $d' \in \Delta^{\mathcal{I}}$ with $\rho^{\mathcal{I}}(\pi(x), d') = R$ and $d' \in D^{\mathcal{I}}$. Create y, set $\pi(y) = d'$. For each z, choose $S_z = \rho^{\mathcal{I}}(\pi(z), d')$. Both invariants are maintained. Composition consistency (CF) holds because the model satisfies the composition table. After ($\forall$)-propagation, invariant (I) is maintained.

Clash-freeness. Invariant (I) prevents concept clashes; invariant (II) ensures composition consistency.

Demand satisfaction. The only concern is blocked nodes. Let x be blocked by y with $\mathcal{L}(x) = \mathcal{L}(y)$, $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, and $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$. Suppose $\exists R.D \in \mathcal{L}(x)$. Then $\exists R.D \in \mathcal{L}(y)$. Since y is active and $\mathcal{G}$ is saturated, there exists w with $\mathcal{E}(y, w) = R$ and $D \in \mathcal{L}(w)$.

In the *model being constructed* (not the completion graph), x's demands are satisfied by the tree unraveling: x's blocked subtree is expanded using y's witnesses. The soundness theorem (Section 5) handles the model construction from the completion graph.

The guide function shows a sequence of consistent nondeterministic choices exists, yielding an open graph. □

7 Computational Evidence for Triangle-Type-Set Stabilization

The termination proof (Theorem 4.1) gives a non-elementary bound. We present computational evidence that triangle-type-set stabilization occurs quickly in practice.

7.1 The PO-incoherent counterexample

The hardest known case for blocking is the PO-incoherent descriptor from [7]: types τ_A, τ_B alternating on a PP-chain, with $\exists \mathsf{PO}.A \in \tau_A$ and $\forall \mathsf{PO}.\neg A \in \tau_B$. Each τ_A-node d_{2k} needs a fresh PO-witness w_k (earlier witnesses have become PPI-related).

7.2 Node-identity profiles never match

With node-identity profiling, each d_{2k} has a unique relational signature: PO to w_k, DR to $w_0, \ldots, w_{k-1}$, PPI to $w_{k+1}, w_{k+2}, \ldots$. Since k grows without bound, no two τ_A-nodes ever share a profile. This was verified computationally [8].

7.3 Abstract triangle-type sets DO match

Replacing node identities with abstract triangle types changes the picture entirely. The script `triangle_type_saturation_check.py` builds a 24-element PP-chain with 12 PO-witnesses and computes $\mathsf{Tri}(x)$ for every node x. Basic Tri-matching stabilizes at $k = 2$:

Node type	Tri stabilizes	Interior range	|Tri|
τ_A	d_4 (k=2)	$d_4 = d_6 = \cdots = d_{18}$	68
τ_B	d_5 (k=2)	$d_5 = d_7 = \cdots = d_{19}$	56
σ	w_2 (k=2)	$w_2 = w_3 = \cdots = w_9$	57

7.4 Tri-neighborhood equivalence also stabilizes

The strengthened condition $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$ requires that the Tri-values among neighbors also match per (relation, type) pair. The script `tri_neighborhood_check.py` verifies this. Result: Tri-neighborhood equivalence stabilizes one step later, at $k = 3$:

Node type	Tri stabilizes	TNbr stabilizes
τ_A	d_4 (k=2)	d_6 (k=3)
τ_B	d_5 (k=2)	d_7 (k=3)
σ	w_2 (k=2)	w_3 (k=3)

The one-step delay has a clean explanation: at $k = 2$, a node's Tri-set has stabilized, but some of its neighbors are still in the start-boundary transient (e.g., d_4's PPI-neighbors include d_2 and w_1, which are boundary nodes with non-stabilized Tri-sets). At $k = 3$, *all* neighbors are also in the stabilized interior, so the TNbr signatures match.

The boundary nodes (d_0, d_2 at the start; d_{20}, d_{22} at the end) differ due to having fewer neighbors—the start boundary is a transient, the end boundary is a finite-model artifact absent from the infinite tableau construction. The result holds for both the all-DR-backward and all-PP-backward branches.

7.5 Why stabilization occurs

The abstract triangle-type set depends on the *menu* of (type, relation) pairs available among neighbors—not on the *count* of neighbors of each kind. Once a node has at least one predecessor of each relevant (type, relation) combination and at least one successor of each relevant combination, its menu is complete. Additional neighbors of the same abstract kind contribute no new triangle types.

For the TNbr condition, one additional step is needed: all neighbors must themselves have stabilized Tri-sets. Since all three node types (τ_A, τ_B, σ) stabilize simultaneously at $k = 2$, a node at depth $k = 3$ has all its neighbors at depth ≥ 2, hence all with stabilized Tri-sets.

8 Discussion

8.1 Resolving the blocking dilemma

Tri-neighborhood blocking occupies a precise intermediate position between the two extremes of the blocking dilemma:

Blocking condition	Terminates?	Correct unraveling?
Type-equality	Always	Not always
Node-identity profile	Not always	Always
Triangle-type-set only	Always	Partial
Tri-neighborhood	**Always**	**Always**

The key insight is that the non-termination argument for node-identity blocking uses *concrete* information (which specific witnesses), while the correctness argument needs only *abstract* information (which relational patterns). Basic triangle-type-set blocking ($\mathsf{Tri}(x) = \mathsf{Tri}(y)$) strips away the concrete information but retains only the *first-person* perspective—what triangle types x itself participates in. The Tri-neighborhood condition (iii) adds the *third-person* perspective: for each neighbor type and relation, the *set of Tri-values* among neighbors must also match. This ensures that when a blocked node's copy appears as a *second or third* vertex in a triangle, the triangle type is still in $T(\mathcal{G})$.

8.2 Relationship to the two-tier quotient

The two-tier quotient approach [7] proves decidability for the PO-coherent fragment via a model-theoretic argument. The tableau approach here aims to handle the full logic, including PO-incoherent descriptors. The two approaches are complementary:

- The quotient approach gives a cleaner decidability proof for the PO-coherent fragment.
- The tableau approach addresses the PO gap by using Tri-neighborhood blocking, which is insensitive to the PO-coherent/incoherent distinction.

8.3 Complexity

The termination bound is non-elementary due to the triple-exponential active-node count (the space of $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples is triply exponential). We do not claim ExpTime membership. An ExpTime bound would require showing that the number of active nodes is at most singly exponential in $|C_0|$—specifically, that Tri-neighborhood stabilization occurs within a number of steps polynomial in $|\mathsf{Tp}(C_0)|$.

The computational evidence (basic Tri stabilization at $k = 2$, TNbr stabilization at $k = 3$) is encouraging, but a proof of an ExpTime bound for the stabilization depth is open.

8.4 Extension to $\mathcal{ALCI}_{RCC8}$

RCC8 has 8 base relations and a more complex composition table. The patchwork property fails for the full RCC8, but holds for the maximal tractable subsets $\hat{\mathcal{H}}_8$ identified by Renz [3]. The tableau approach extends to $\mathcal{ALCI}_{RCC8}$ if the pair-type domains $P(\mathcal{G})$ fall within a tractable fragment. This is a non-trivial condition that we do not address here.

8.5 Honest assessment of gaps

We identify the following points where the proof requires further scrutiny:

1. **Intra-subtree T-closure (Lemma 5.6).** The proof that the map-based assignment $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$ is T-closed works cleanly for cross-subtree pairs (where n_1, n_2 are

distinct nodes in $\mathcal{G}$). For intra-subtree pairs where the same tableau node appears at multiple positions in the unraveling (due to blocked copies), the argument relies on all three blocking conditions: (i) $\mathcal{L}(w) = \mathcal{L}(\beta(w))$, (ii) $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$, and (iii) $\mathsf{TNbr}(w) = \mathsf{TNbr}(\beta(w))$. Condition (ii) handles triangles where the blocked node is the *first* vertex (its own perspective). Condition (iii) handles the harder case where the blocked node appears as a *second or third* vertex: when element d maps to a node b that is a neighbor of w, the triangle type involving d must be in $T(\mathcal{G})$. Since $\mathsf{TNbr}(w) = \mathsf{TNbr}(\beta(w))$, for each neighbor b of w with relation R and type τ, there exists a neighbor b' of $\beta(w)$ with the same relation and type *and* $\mathsf{Tri}(b) = \mathsf{Tri}(b')$, which ensures the triangle types involving b's perspective are also preserved.

The remaining edge case is when $n_1 = n_2$ (both copies map to the same node), making $\rho(d_1, d_2)$ undefined. This is resolved by assigning $\rho(d_1, d_2) \in \mathsf{Safe}(\mathsf{tp}(d_1), \mathsf{tp}(d_2))$; since $\mathsf{tp}(d_1) = \mathsf{tp}(d_2) = \mathcal{L}(n)$ and the type is realizable, such a safe relation exists.

2. **Monotonicity of triangle-type sets.** The termination proof claims triangle-type sets grow monotonically. This is true for node addition (new triples $\Rightarrow$ potentially new triangle types). For label changes (via $\forall$-propagation), a concept added to $\mathcal{L}(x)$ changes $\mathcal{L}(x)$ to a superset, which changes the Hintikka-type component in all triangle types involving x. Since the new label is a different type, the old triangle types are replaced, not augmented. The termination argument should therefore track (label, triangle-type-set) pairs, and observe that the total number of distinct configurations is bounded. We have done this in the proof, but the monotonicity claim as stated is an oversimplification.

3. **Stabilization depth.** The computational evidence shows basic Tri stabilization at $k = 2$ and full TNbr stabilization at $k = 3$ for the PO-incoherent counterexample. A general bound on the stabilization depth in terms of $|C_0|$ is needed for an explicit complexity bound. We conjecture that the stabilization depth is at most $|\mathsf{Tp}(C_0)|$ (the number of distinct types), but do not prove this.

9 Conclusion

We have presented a tableau calculus for $\mathcal{ALCI}_{RCC5}$ with *Tri-neighborhood blocking*: a node x is blocked by an earlier node y when (i) $\mathcal{L}(x) = \mathcal{L}(y)$, (ii) $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, and (iii) $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$. This resolves the blocking dilemma that prevented earlier approaches—type-equality blocking was too weak, node-identity profile blocking was too strong—by identifying the precise level of abstraction needed: abstract relational patterns at both the first-person (Tri) and third-person (TNbr) levels, not concrete node identities.

The calculus is *terminating* (the space of $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples is finite), *sound* (model construction via triangle-type-filtered unraveling, with global consistency from full RCC5 tractability), and *complete* (a model guides all nondeterministic choices). Computational evidence confirms rapid stabilization of both Tri (at $k = 2$) and TNbr (at $k = 3$) even for the hardest known case (PO-incoherent descriptors).

If the termination bound can be tightened to singly exponential (which the computational evidence suggests is possible), this yields an ExpTime decision procedure for $\mathcal{ALCI}_{RCC5}$, matching the known PSPACE lower bound [1] up to one exponential.

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